📊 Task 4: Classification Models, Evaluation Metrics & Handling Imbalanced Data

1. Objective & Scenario Analysis

The purpose of this evaluation is to build a robust binary classification pipeline using the Breast Cancer dataset to cleanly distinguish between Malignant (Class 0) and Benign (Class 1) clinical samples. In medical diagnosis workflows, predictive accuracy alone is a highly misleading metric. For example, if a dataset contains 95% benign samples, a naive model that predicts "benign" every single time will achieve a 95% accuracy score while failing to detect a single malignant tumor, leading to critical real-world failure.

2. Metric Selection Justification

To measure performance reliably, the evaluation relies on a structured Confusion Matrix and specific class-level metrics:

- **Confusion Matrix Breakdown:** The baseline model correctly isolated 41 True Negatives (Malignant) and 71 True Positives (Benign). It incurred 1 False Positive and 1 False Negative.
- **The Importance of Recall:** In healthcare diagnostics, **Recall (Sensitivity)** is prioritized over Precision. A low recall score means the model is generating a high number of False Negatives—meaning patients with malignant tumors are incorrectly told they are clear, delaying lifesaving treatment.
- **F1-Score Utility:** The F1-Score calculates the harmonic mean of Precision and Recall. It provides a balanced indicator of model stability when dealing with imbalanced datasets where one class outnumbers the other.

3. Class Imbalance Handling

To protect the model against structural class frequency skewness, an optimized variant of Logistic Regression was implemented using the class_weight="balanced" hyperparameter constraint. This technique dynamically scales the loss function penalty inversely proportional to the class frequencies in the training data, ensuring the minority class exerts a matching influence on weight convergence.

4. Final Model Decision & Comparison

The experimental performance metrics across the different architectures are summarized below:

- **Baseline Logistic Regression:** Accuracy: 98% | Malignant Recall: 98% | Benign Recall: 99% | ROC-AUC: ~0.99

- **Balanced Logistic Regression:** Accuracy: 98% | Malignant Recall: 98% | Benign Recall: 99% | ROC-AUC: ~0.99
- **Decision Tree Classifier:** Accuracy: ~95% | Malignant Recall: ~93% | Benign Recall: ~96% | ROC-AUC: N/A

Justification: Logistic Regression is chosen as the final model for production deployment. It significantly outperformed the Decision Tree across all metrics. Furthermore, the model achieved a near-perfect **ROC-AUC score**, proving exceptional diagnostic discrimination power. Its smooth, probabilistic decision boundary provides continuous risk scores that are more stable and interpretable for medical classification workflows than a decision tree's rigid, hard splits.